

# Proof of the Asymptotic Portfolio Loss Formula in the Basel II ASRF Framework

Why the portfolio loss percentile equals  
the conditional expected loss

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## 1 Introduction

The Basel II regulatory capital framework employs the *Asymptotic Single Risk Factor* (ASRF) model, derived from Vasicek’s portfolio credit risk model, to determine capital requirements for credit risk. The regulatory formula computes the loss at a given confidence level as the *conditional expected loss* of the portfolio, evaluated at a suitably adverse realization of a single systematic risk factor.

The equivalence between these two concepts —the simulated portfolio loss percentile and the conditional expected loss— is far from obvious. This note presents a step-by-step proof, building on Proposition 2.5.4 of Bluhm, Overbeck, and Wagner<sup>1</sup>, with the aim of making each logical step accessible to risk practitioners who may not work with these derivations on a daily basis.

The proof rests on three pillars: (i) conditional independence of defaults given the systematic factor, (ii) the conditional law of large numbers that eliminates idiosyncratic risk in granular portfolios, and (iii) the monotonicity of the conditional loss function, which maps quantiles of the factor directly onto quantiles of the loss distribution.

## 2 The single-factor model

**Definition 1** (ASRF model). Consider a portfolio of  $N$  credits. For each obligor  $i = 1, \dots, N$ , the latent variable (*asset value*) is modeled as:

$$r_i = w_i \cdot Z + \sqrt{1 - w_i^2} \varepsilon_i \quad (1)$$

where:

- $Z \sim N(0, 1)$  is the **systematic factor**, common to all credits.
- $\varepsilon_i \sim N(0, 1)$  is the **idiosyncratic factor** for credit  $i$ .
- $w_i = \sqrt{\rho_i} \in (0, 1)$  is the systematic factor loading, where  $\rho_i$  is the asset correlation.
- $Z$  and the  $\varepsilon_i$  are mutually independent.
- $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$  for  $i \neq j$ .

From the structure (1) it follows immediately that  $r_i \sim N(0, 1)$  and  $\text{Cov}(r_i, r_j) = w_i w_j$  for  $i \neq j$ . Under the homogeneous correlation assumption used by Basel II for residential mortgages ( $w_i = w = \sqrt{0.15}$  for all  $i$ ), the pairwise correlation between any two credits is exactly  $\rho = 0.15$ .

**Definition 2** (Default and loss). Credit  $i$  defaults when its latent variable falls below a threshold  $d_i$ :

$$Y_i = \mathbf{1}(r_i < d_i) \quad (2)$$

Since  $r_i \sim N(0, 1)$ , the probability of default is  $PD_i = \Phi(d_i)$ , and therefore  $d_i = \Phi^{-1}(PD_i)$ .

The **total portfolio loss** is:

$$L_N = \sum_{i=1}^N \ell_i \cdot Y_i, \quad \text{where } \ell_i = LGD_i \cdot EAD_i \quad (3)$$

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<sup>1</sup>C. Bluhm, L. Overbeck, and Ch. Wagner. *An Introduction to Credit Risk Modelling*. Chapman & Hall/CRC, 2003, pp. 55–91.

The connection to the Merton model<sup>2</sup> is worth noting:  $r_i$  can be interpreted as the (standardized) asset return of obligor  $i$ ,  $Z$  captures economy-wide fluctuations,  $\varepsilon_i$  represents firm-specific shocks, and  $d_i$  is the distance-to-default threshold. When  $r_i$  falls below  $d_i$ , the asset value is insufficient to cover the obligation, triggering default.

### 3 Conditional independence

This result is the cornerstone of the entire derivation.

**Lemma 1** (Conditional independence). *Given  $Z = z$ , the default indicators  $Y_1, Y_2, \dots, Y_N$  are mutually independent.*

*Intuition.* In the model, the *only* source of dependence between credit  $i$  and credit  $j$  is the common factor  $Z$ . Once the state of the economy is known ( $Z = z$ ), each credit depends solely on its own idiosyncratic shock  $\varepsilon_i$ , and these are independent by construction. The systematic factor acts as a “switch”: it creates correlation unconditionally, but once observed, it releases each credit to behave independently.

*Proof.* Given  $Z = z$ , the latent variable of credit  $i$  reduces to:

$$r_i \mid Z = z = w_i z + \sqrt{1 - w_i^2} \varepsilon_i$$

which is a function of  $\varepsilon_i$  alone (since  $w_i z$  is a constant, conditionally). Because the  $\varepsilon_i$  are mutually independent by hypothesis, the random variables  $r_i \mid Z = z$  are mutually independent. Consequently, the events  $\{r_i < d_i\}$ —that is, the default indicators  $Y_i$ —are conditionally independent given  $Z$ .  $\square$

### 4 Conditional probability of default

**Lemma 2** (Conditional PD). *The probability of default of credit  $i$ , conditional on  $Z = z$ , is:*

$$p_i(z) := \mathbb{P}(Y_i = 1 \mid Z = z) = \Phi\left(\frac{d_i - w_i z}{\sqrt{1 - w_i^2}}\right) \quad (4)$$

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<sup>2</sup>R.C. Merton. On the pricing of corporate debt: The risk structure of interest rates. *Journal of Finance*, 29:449–470, 1974.

*Proof.*

$$\begin{aligned}
\mathbb{P}(Y_i = 1 \mid Z = z) &= \mathbb{P}(r_i < d_i \mid Z = z) \\
&= \mathbb{P}\left(w_i z + \sqrt{1 - w_i^2} \varepsilon_i < d_i\right) \\
&= \mathbb{P}\left(\varepsilon_i < \frac{d_i - w_i z}{\sqrt{1 - w_i^2}}\right) \\
&= \Phi\left(\frac{d_i - w_i z}{\sqrt{1 - w_i^2}}\right)
\end{aligned}$$

where the last step uses  $\varepsilon_i \sim N(0, 1)$ . □

*Remark.* The function  $p_i(z)$  is **strictly decreasing** in  $z$ : as  $z$  decreases (adverse economic scenario), the probability of default increases. This is consistent with economic intuition — a negative systematic shock deteriorates the credit quality of the entire portfolio. Conversely,  $p_i(z) \rightarrow 0$  as  $z \rightarrow +\infty$  (benign scenario) and  $p_i(z) \rightarrow 1$  as  $z \rightarrow -\infty$  (catastrophic scenario).

## 5 Conditional expected loss

**Definition 3** (Conditional expected loss function). We define:

$$\mu_N(z) := \mathbb{E}[L_N \mid Z = z] = \sum_{i=1}^N \ell_i \cdot p_i(z) \quad (5)$$

This follows directly from the linearity of expectation and the fact that, given  $Z = z$ , each  $Y_i$  is Bernoulli with parameter  $p_i(z)$ .

*Remark* (Key properties of  $\mu_N(z)$ ). The function  $\mu_N(z)$  is:

1. **Continuous** in  $z$  (composition of continuous functions).
2. **Strictly decreasing** in  $z$  (sum of strictly decreasing functions with positive weights  $\ell_i > 0$ ).
3. **Bounded**:  $0 \leq \mu_N(z) \leq \sum_i \ell_i$  for all  $z$ .

Strict monotonicity implies that  $\mu_N$  is **invertible**: there is a one-to-one correspondence between values of the systematic factor and conditional expected loss levels. This invertibility is essential for the final step of the proof.

## 6 The conditional law of large numbers

This is the central technical result: in a sufficiently granular portfolio, the realized loss converges to the conditional expected loss.

## 6.1 Granularity condition

**Definition 4** (Granular portfolio). A portfolio satisfies the **granularity condition** if no single exposure dominates the total loss. Formally:

$$H_N := \frac{\sum_{i=1}^N \ell_i^2}{\left(\sum_{i=1}^N \ell_i\right)^2} \xrightarrow{N \rightarrow \infty} 0 \quad (6)$$

The index  $H_N$  is analogous to the Herfindahl-Hirschman Index of concentration. The condition is satisfied, for instance, when exposures are identical ( $\ell_i = \ell$  for all  $i$ , giving  $H_N = 1/N \rightarrow 0$ ) or, more generally, when individual exposures are uniformly bounded while the total exposure grows without bound.

For residential mortgage portfolios—which typically contain thousands or tens of thousands of loans of comparable size—the granularity condition is naturally satisfied. This is one of the reasons why the ASRF model is particularly well suited for mortgage insurance and housing credit portfolios.

## 6.2 Convergence result

**Proposition 1** (Convergence of portfolio loss — Proposition 2.5.4). *Under the granularity condition (6), for every  $z \in \mathbb{R}$ :*

$$L_N \xrightarrow{P|Z=z} \mu_N(z) \quad \text{as } N \rightarrow \infty$$

*That is, conditional on any realization of the systematic factor, the portfolio loss converges in probability to its conditional expectation.*

*Intuition.* Fix  $Z = z$  — freeze the economy in a particular state. Now we have  $N$  independent random variables  $Y_i$  (Bernoulli with parameters  $p_i(z)$ ), each weighted by  $\ell_i$ . If the portfolio is granular (no single credit dominates), the weighted sum behaves like an average of many independent variables. By the law of large numbers, the outcome concentrates around its expectation. It is the same principle that makes the frequency of heads converge to 1/2 after many coin tosses: *idiosyncratic uncertainty is diversified away*.

*Proof.* Fix  $Z = z$  and compute the conditional variance of  $L_N$ . By Lemma 1, the  $Y_i$  are conditionally independent, so:

$$\text{Var}(L_N | Z = z) = \sum_{i=1}^N \ell_i^2 \cdot \underbrace{\text{Var}(Y_i | Z = z)}_{= p_i(z)(1-p_i(z))} \quad (7)$$

Using the elementary bound  $p(1-p) \leq \frac{1}{4}$  for all  $p \in [0, 1]$ :

$$\text{Var}(L_N | Z = z) \leq \frac{1}{4} \sum_{i=1}^N \ell_i^2 \quad (8)$$

Now apply **Chebyshev's inequality**. For any  $\delta > 0$ :

$$\begin{aligned} \mathbb{P}(|L_N - \mu_N(z)| > \delta \mid Z = z) &\leq \frac{\text{Var}(L_N \mid Z = z)}{\delta^2} \\ &\leq \frac{1}{4\delta^2} \sum_{i=1}^N \ell_i^2 \end{aligned} \quad (9)$$

Under the granularity condition, as  $N \rightarrow \infty$  with the total exposure  $S_N = \sum_i \ell_i$  growing, the concentration index satisfies  $H_N = \sum_i \ell_i^2 / S_N^2 \rightarrow 0$ , which implies  $\sum_i \ell_i^2 / \delta^2 \rightarrow 0$  for any fixed  $\delta$  relative to the portfolio scale<sup>3</sup>. Therefore:

$$\mathbb{P}(|L_N - \mu_N(z)| > \delta \mid Z = z) \xrightarrow{N \rightarrow \infty} 0 \quad \forall \delta > 0$$

This is precisely the convergence in probability of  $L_N$  to  $\mu_N(z)$ , conditional on  $Z = z$ .  $\square$

*Remark* (Interpretation). The result states that, for a granular portfolio, **idiosyncratic risk is fully diversified away**. Once the state of the economy is known ( $Z = z$ ), the portfolio loss is essentially determined — it is  $\mu_N(z)$ , with no material residual variability. All remaining uncertainty in  $L_N$  comes from *not knowing which value  $Z$  will take*, i.e., from systematic risk alone.

## 7 The loss percentile as a conditional expectation

We now connect the convergence result to the Basel formula.

**Proposition 2** (Percentile–conditional expectation equivalence). *Under the granularity condition, as  $N \rightarrow \infty$ :*

$$\text{VaR}_\alpha(L_N) \longrightarrow \mu_N(\Phi^{-1}(1 - \alpha)) = \mathbb{E}[L_N \mid Z = \Phi^{-1}(1 - \alpha)] \quad (10)$$

*Intuition.* By Proposition 1, the loss  $L_N$  behaves essentially as the deterministic function  $\mu_N(Z)$  applied to the random factor  $Z$ . Since  $\mu_N$  is strictly decreasing (Remark 5), high losses (the right tail of  $L_N$ ) correspond to low values of  $Z$  (the left tail: adverse economy). Finding the  $\alpha$ -percentile of  $L_N$  is therefore equivalent to evaluating  $\mu_N$  at the  $(1 - \alpha)$ -percentile of  $Z$ .

The flip from  $\alpha$  to  $(1 - \alpha)$  comes from the decreasing monotonicity: we want the point where  $P(L_N \leq l) = \alpha$ , but since loss decreases when  $Z$  increases, this corresponds to  $P(Z \geq z^*) = \alpha$ , i.e.,  $P(Z \leq z^*) = 1 - \alpha$ .

*Proof.* By definition, the Value at Risk at level  $\alpha$  is:

$$\text{VaR}_\alpha(L_N) = \inf\{l \in \mathbb{R} : \mathbb{P}(L_N \leq l) \geq \alpha\}$$

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<sup>3</sup>More precisely, the bound (9) involves  $\sum \ell_i^2$  in absolute terms. For convergence of the non-normalized loss, we require that  $\sum \ell_i^2 \rightarrow 0$  or, equivalently, that  $S_N \rightarrow \infty$  and  $H_N \rightarrow 0$  simultaneously. This holds in real mortgage portfolios where  $N$  grows and exposures remain bounded.

**Step 1.** By Proposition 1, in the limit  $N \rightarrow \infty$ ,  $L_N$  converges to  $\mu_N(Z)$ . Therefore:

$$\mathbb{P}(L_N \leq l) \longrightarrow \mathbb{P}(\mu_N(Z) \leq l) \quad (11)$$

**Step 2.** Since  $\mu_N$  is strictly decreasing, the inequality reverses when applying  $\mu_N^{-1}$ :

$$\mathbb{P}(\mu_N(Z) \leq l) = \mathbb{P}(Z \geq \mu_N^{-1}(l)) = 1 - \Phi(\mu_N^{-1}(l)) \quad (12)$$

**Step 3.** Setting equal to  $\alpha$  and solving for  $l$ :

$$\begin{aligned} 1 - \Phi(\mu_N^{-1}(l)) &= \alpha \\ \Phi(\mu_N^{-1}(l)) &= 1 - \alpha \\ \mu_N^{-1}(l) &= \Phi^{-1}(1 - \alpha) \\ l &= \mu_N(\Phi^{-1}(1 - \alpha)) \end{aligned}$$

**Step 4.** Substituting the definition of  $\mu_N$  (5) and  $p_i$  (4), with  $z^* = \Phi^{-1}(1 - \alpha) = -\Phi^{-1}(\alpha)$ :

$$\begin{aligned} VaR_\alpha(L_N) &= \sum_{i=1}^N \ell_i \cdot p_i(\Phi^{-1}(1 - \alpha)) \\ &= \sum_{i=1}^N \ell_i \cdot \Phi\left(\frac{d_i - w_i \cdot \Phi^{-1}(1 - \alpha)}{\sqrt{1 - w_i^2}}\right) \\ &= \sum_{i=1}^N \ell_i \cdot \Phi\left(\frac{d_i + w_i \cdot \Phi^{-1}(\alpha)}{\sqrt{1 - w_i^2}}\right) \end{aligned} \quad (13)$$

where the last step uses  $-\Phi^{-1}(1 - \alpha) = \Phi^{-1}(\alpha)$ .

Replacing  $d_i = \Phi^{-1}(PD_i)$  and  $\ell_i = LGD_i \cdot EAD_i$ :

$$\boxed{VaR_\alpha(L_N) = \sum_{i=1}^N LGD_i \cdot EAD_i \cdot \Phi\left(\frac{\Phi^{-1}(PD_i) + \sqrt{\rho} \Phi^{-1}(\alpha)}{\sqrt{1 - \rho}}\right)} \quad (14)$$

□

Equation (14) is the **Basel II IRB risk weight formula**<sup>4</sup>. The proof reveals that this formula is not an ad hoc approximation: it is the *exact* asymptotic result of a portfolio credit risk model with a single systematic factor, under the granularity assumption. Its applicability to mortgage portfolios — which are inherently granular, consisting of thousands of relatively homogeneous exposures — is a natural consequence of this structure.

The additive form of (14) (the portfolio VaR equals the sum of individual contributions) is itself a remarkable property: it means that each credit's contribution to the portfolio risk can be computed independently, enabling the “building-block” approach to regulatory capital that makes Basel II operationally tractable.

<sup>4</sup>Basel Committee on Banking Supervision. *An Explanatory Note on the Basel II IRB Risk Weight Functions*. Bank for International Settlements, 2005.

## References

- [1] C. Bluhm, L. Overbeck, and Ch. Wagner. *An Introduction to Credit Risk Modelling*. Chapman & Hall/CRC, 2003. Proposition 2.5.4, pp. 55–91.
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- [4] Basel Committee on Banking Supervision. *An Explanatory Note on the Basel II IRB Risk Weight Functions*. Bank for International Settlements, 2005.